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A scientist identified and counted the invertebrate species present in samples taken at two sites in a river. The scientist also measured the rate of water flow at each site.

His results are shown in **Table 2** and **Table 3**.

Table 2

Invertebrate species	Site 1	Site 2
Anglers' Curse mayfly	17	5
Flat-headed mayfly	6	8
Slate Drake mayfly	0	6
Water beetle	12	13
Midge fly	13	13
Total number caught	48	45

Table 3

	Site 1	Site 2
Index of diversity		4.7
Rate of water flow / cm s ⁻¹	1–14	30–60

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Complete **Table 3** by calculating the index of diversity (d) at **Site 1**.

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

[1 mark]

Index of diversity (d) = _____



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Explain why it is more useful to calculate an index of diversity than to record species richness.

[2 marks]

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Suggest how the scientist measured the rate of water flow in the river.

[1 mark]

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Use information in **Table 2** and **Table 3** to suggest and explain a reason for the difference in the numbers of Slate Drake mayfly at these sites in this river.

[2 marks]

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It was important that the sampling procedure was standardised when collecting the Slate Drake mayflies from the two sites.

Give **one** way in which the sampling procedure could be standardised.

[1 mark]

7

Turn over ►

